

DSA Pattern Sheet: Binary Search on Answer (Threshold / Leveling Pattern)

Pattern Name

Binary Search on Answer (Threshold / Leveling Pattern)

Also combines:

- Greedy
 - Arithmetic Progression (Range Sum)
 - Leveling Technique
-

Core Idea

Most beginners think:

"Which operation should I perform next?"

Experts think:

"What is the final threshold (level) after all operations?"

Instead of simulating every operation, search for the **final answer directly**.

Thinking Evolution

Brute Force

↓

Simulate every operation

↓

Too Slow

↓

Can multiple operations be grouped together?

↓

Represent grouped operations by a Threshold / Level

↓

Can we check any threshold in $O(n)$?

↓

YES

↓

Is the check monotonic?

↓

YES

↓

Binary Search on Answer

↓

Compute final result mathematically

Recognition Checklist

Whenever you see

- ✓ Repeated operations
- ✓ Large constraints (10^5 or more)
- ✓ Simulation is expensive
- ✓ Values increase/decrease gradually
- ✓ Need Maximum/Minimum answer

Ask yourself

Can I stop simulating
and instead search for a final level?

If YES

↓

Write a **check(mid)** function.

If

```
check(mid)
```

is monotonic

↓

Binary Search on Answer.

What is a Threshold?

A threshold is simply a **common stopping point**.

Example

```
10  
7  
5
```

Suppose threshold

```
6
```

Everything above 6 becomes

```
6
```

Result

```
6  
6  
5
```

We no longer care about individual operations.

We only care about

"How much work is needed to reach this level?"

How to Build the Logic

Step 1

Start with brute force.

Ask

```
What am I repeating?
```

Example

```
Sell one ticket.  
Sell one ticket.  
Sell one ticket.
```

Step 2

Ask

```
Can many operations be done together?
```

Example

Instead of

```
10  
9  
8  
7
```

Calculate

```
10+9+8+7
```

using Arithmetic Progression.

Step 3

Ask

```
Where should I stop?
```

Instead of thinking

```
Next ticket
```

Think

```
Final Level
```

Step 4

Create a Check Function

Example

```
Threshold = mid
```

Question

```
If every value above mid is reduced to mid,  
How many operations are needed?
```

Usually

```
O(n)
```

Step 5

Observe the Behaviour

As threshold increases

Operations decrease

or

As threshold decreases

Operations increase

This creates

Monotonic Behaviour

which is the key requirement for Binary Search.

Monotonic Property

Example

10
7
5

Threshold	Operations
8	2
7	3
6	5
5	7

Observe

Threshold ↑

Operations ↓

Always.

Whenever this happens

↓

Binary Search.

General Binary Search Template

```
low = minimum_answer
high = maximum_answer

while low <= high:

    mid = (low + high) // 2

    if check(mid):

        answer = mid

        # move according to problem
        low = mid + 1

    else:

        high = mid - 1
```

Entire trick lies inside

```
check(mid)
```

Generic Check Function

Most Binary Search on Answer problems become

```
def check(mid):  
    value = 0  
    for x in arr:  
        # Compute work for this threshold  
    return value >= required
```

Sometimes

$\geq$

Sometimes

$\leq$

depends on the problem.

Arithmetic Progression Trick

Whenever simulation looks like

```
10  
9  
8  
7
```

Instead use


```
First = 10

Last = 7

Count = 4

Sum = (First + Last) × Count // 2
```

Converts

```
O(k)

↓

O(1)
```

Common Patterns That Use This Idea

1. Binary Search on Answer

Examples

- Koko Eating Bananas
- Capacity To Ship Packages
- Allocate Minimum Pages
- Painter Partition
- Aggressive Cows
- Maximum Candies Allocation

2. Threshold / Leveling

Examples

```
Reduce heights

Fill water

Cut wood

Allocate capacity

Sell decreasing values
```

Equalize values

3. Greedy

Local optimal choice

Always choose best current option.

Example

Sell highest ticket.

4. Arithmetic Progression

Whenever values are

10

9

8

7

Don't loop.

Use formula.

Common Interview Flow

Interviewer gives

Brute Force

↓

Improve

Heap

↓

Still slow

↓

Observe repeated pattern

↓

Batch operations

↓

Threshold

↓

Binary Search

↓

Math Formula

Real World Examples

Water Tanks

10L

7L

5L

Lower every tank to

6L

instead of removing water litre by litre.

Salary Reduction

Employees

100

90

70

Reduce everyone above

80

No need to reduce salary one dollar at a time.

Mountain Leveling

10

8

6

Bulldoze everything above height

6

instead of removing one rock at a time.

Ticket Selling Problem

Instead of

Sell

10

9

8

7

Think

Reduce seller

10

↓

6

Then compute earnings directly.

Pattern Recognition Sheet

Whenever you see

Repeated simulation

↓

Large constraints

↓

Need Max/Min answer

↓

Can imagine a common level

↓

Can write check(mid)

↓

Check is monotonic

↓

Binary Search on Answer

Questions to Ask Yourself

1. What is being repeated?
 2. Can I batch these operations?
 3. Can I replace operations by a threshold?
 4. Can I check any threshold in $O(n)$?
 5. Does the answer change monotonically?
 6. Can Binary Search find this threshold?
 7. Can the remaining work be computed using mathematics instead of simulation?
-

Golden Rules

- ✓ Simulating one operation at a time is often too slow.
 - ✓ Think in **Levels**, not individual operations.
 - ✓ Threshold problems usually become Binary Search.
 - ✓ Consecutive values usually indicate Arithmetic Progression.
 - ✓ Binary Search on Answer is about searching a **value**, not an **index**.
-

Master Template

Repeated Simulation

↓

Identify expensive operation

↓

Can operations be grouped?

↓

Represent group using Threshold

↓

Build `check(mid)`

↓

Check is Monotonic

↓

Binary Search

↓

Mathematical Computation

↓

Optimal Solution

One-Line Memory Trick

"When simulation is too slow, stop thinking about the next operation—start thinking about the final threshold."